

Trading to Success: How Front Office Strategy Shapes Regular Season Performance in the WHL

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Abstract—In this paper, we model Western Hockey League (WHL) goals for, goals against, and win totals during the regular season using random walks. Our approach highlights that goal differential fails to explain the variance in team records. To improve our predictions, we generate player and team ratings, using regression and random forest models to evaluate trades impacts and front office strategy. These trades, often made as teams sacrifice draft picks for Memorial Cup runs, influence scoring parameters and Elo ratings, impacting regular season outcomes. We hypothesize mid-season trades significantly affect results as teams acquire top players before the deadline.

I. INTRODUCTION

FOUNDED in 1966, the Western Hockey League (WHL) is one of the best developmental hockey circuits in the world. Alongside its Canadian Hockey League (CHL) counterparts, the Ontario Hockey League (OHL) and Quebec Maritimes Junior Hockey League (QMJHL), the WHL holds a major junior designation, placing it at the peak of Canadian junior hockey. With 22 teams spread across 4 Canadian provinces and 2 U.S. states, the WHL boasts over 127 players named to National Hockey League (NHL) opening night rosters for the 2024-2025 season. The league is a junior hockey powerhouse and the pinnacle of the sport for skaters older than 16 and younger than 20.

A. League Mechanics

In 2018, the WHL reduced the regular season from 72 games to 68 games. Each of the two WHL conferences, Eastern and Western, consists of 11 teams. Teams in the Eastern Conference play one regular season game against every team in the Western Conference. The Western Conference is divided into the U.S. Division (6 teams) and the B.C. Division (5 teams), while the Eastern Conference is split into the Central Division (5 teams) and the Eastern Division (6 teams). The number of games a team plays against divisional opponents varies depending on the size of their division, but the total number of in-conference games for each team is 57. Therefore, the 68 game season is broken into 11 inter-conference games and 57 in-conference games.¹

Winning the Western Hockey League (WHL) grants franchises a bid to the Memorial Cup, the premier tournament in the Canadian Hockey League (CHL). Due to the CHL's age cap on players, frequent transactions occur between franchises leading up to the January trade deadline. Competitive teams

typically seek top-tier talent to bolster their Memorial Cup hopes, while lower-ranked teams may give up their best players for draft picks or younger prospects. In theory, the expected value of a top finish in the WHL should follow a cyclical pattern, as teams adjust their rosters for both short- and long-term success.

B. Model Motivation

If front office strategies remained consistent, the five-year eligibility window for CHL players would suggest that teams maximize their chances of winning the WHL and qualifying for the Memorial Cup Playoffs approximately every 2.5 to 3 years. However, this is not the case. In the WHL, the Saskatoon Blades have remained atop the Eastern Conference standings since the COVID pandemic, while the Regina Pats have not finished higher than fourth in the Eastern Division of the Eastern Conference (six teams) since the 2017-2018 season. The Moose Jaw Warriors represent the middle ground: they won the WHL in 2024 but sacrificed many future draft picks to do so. As of the first half of the 2025 season, they are ranked last in the Eastern Conference.

Thus, we hypothesize that front office strategy plays a significant role in WHL team success. While program privacy restrictions prevent us from developing a direct metric of scouting effectiveness or financial investment, we can evaluate front office decisions through the lens of Elo scores. In this paper, we construct comprehensive datasets that include player and goalie statistics, head-to-head matchups, team performance metrics, and a trade database spanning the last six non-COVID seasons. Using these datasets, we develop two variations of a random walk model: one based on a Gaussian random step for goal differential (using the mean goal differential and its variation across games in a season), and the other employing a Poisson distribution approach. After confirming that the random walk model cannot effectively predict win totals on its own, we extend the analysis to evaluate the impact of player trades and front office decisions.

We understand that a player will perform differently in different team environments. As such, modeling each player's impact on their new franchise will provide a better model than leveraging their statistics independent of their team. This approach also allows for a method to compare goalies to skaters, as their statistics cannot be directly compared. Moreover, a top defenseman may have very few points compared to a top forward, but if they can lower their team's goals-against average,

¹Please note the first two contextual paragraphs were initialized in our last Mini Project. We feel this provides necessary context about the WHL.

their value might be understated. By calculating each team's net Elo score after acquiring or losing a player, we define a metric through which front office impact can be quantified. The ultimate goal of this study is to assess whether integrating trade data improves the model's predictive accuracy, providing deeper insights into how player movements influence overall team performance.

II. DATASET OVERVIEW

With our datasets, we have access to team statistics since 2012-2013 season, game-by-game player statistics since the 2012-2013 season, and trade statistics since the 2017-2018 season. All datasets have been linked to a Kaggle page in Appendix C. Explanation of basic hockey metrics have been included in Appendix D.

A. Team Statistics

Team statistics include, but are not limited to: total goals, total goals against, wins, losses, total penalty minutes, end-of-regular season conference rank, games played, and various per-game statistics extrapolated from the aforementioned metrics.

B. Player Statistics

The player statistics database is more straightforward. For skaters, we track goals, assists, plus-minus, and shots on goal, while for goalies, we record goals against, saves, and game results (e.g., win, loss, overtime loss, shootout loss). These metrics are compiled on a game-by-game basis, allowing us to quantify player performance both before and after trades.

C. Trade Statistics

The trade database details all WHL transactions. Indexed by transaction date and team ids, every additional column in the dataframe contains an individual player's name or draft pick information.

D. Data Summary

With team, player, and trade statistics in place, we can develop our model for front office strategy and trade optimality. Before proceeding, we first assess whether we can accurately predict a team's record using a goal differential-driven random walk model. If the model holds, it will suggest that the impact of trades within specific time windows is less significant than strengthening season-wide aggregate statistics, as the base regular season metrics would serve as a strong predictor.

III. RANDOM WALK RUDIMENTARY MODEL

A random walk is a stochastic process defined as $X_n = X_{n-1} + \xi_n$, where X_0 is the initial position, and ξ_n are independent and identically distributed random variables representing the step sizes and n is the discrete time step. A hockey game can be modeled as a random walk with step sizes dictated by two distinctive random variables: Gaussian and Poisson. In our first formulation, we try to predict a

team's end-of-season goal differential by leveraging their average goals for and average goals against. By counting all instances where a team's projected goal differential is positive, we determine their projected regulation win count.² In our next random walk formulation, we will account for their opponents' statistics on a game-by-game basis.

- 1) **Gaussian:** A normal distribution centered around the mean of a team's score differential (calculated as expected goals for minus expected goals against), with a standard deviation that reflects the variability in the team's point differential: $\xi_n \sim \mathcal{N}(\mu, \sigma^2)$.
- 2) **Poisson:** A discrete probability distribution that models the number of events (e.g., goals scored) occurring within a fixed interval of time or space, assuming the events happen independently and at a constant average rate: $\xi_n \sim \text{Poisson}(\lambda)$.

To accurately represent a 68-game season in the WHL, we will track the cumulative goal differential and the number of wins. Given that many teams have comparable offensive and defensive statistics in aggregate, our objective will not be to predict every individual game result with precision. Instead, we will focus on modeling the total win count as accurately as possible.

The Poisson distribution provides a robust framework for modeling scoring patterns in hockey. In the WHL, teams may encounter strong goaltenders or experience periods of poor offensive performance, possibly going scoreless for two consecutive games. However, if the circumstances shift in their favor, a team could score five or more goals in a single period, which accounts for just one-third of a full game. The Poisson distribution is suited for modeling the number of events occurring within a fixed interval. A plot of the Regina Pats' most recent full season projection, along with their random walk trajectories for goals scored, goals conceded, and goal differential, has been shared (Fig 1).

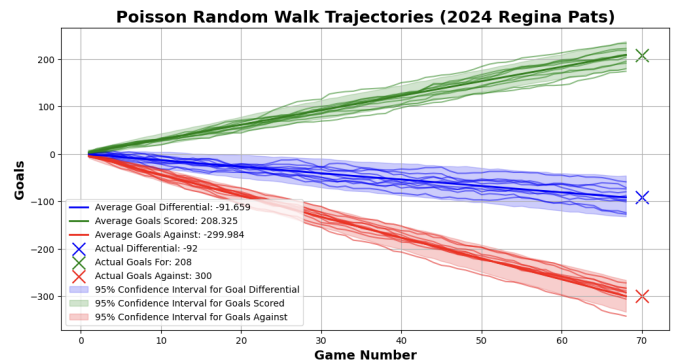


Fig. 1. Random Walk Results for Poisson Distribution: 2024 Regina Pats

²For our random walk model, we will not predict overtime and shootout losses, just overall losses

The Gaussian distribution, on the other hand, enables us to incorporate the spread of goals scored using the σ parameter. It models the dispersion of goals in a more controlled manner than the Poisson distribution and can be manipulated to more accurately reflect the end-of-season cumulative totals. In the Gaussian model (Fig 2), we have scaled each team's goal spread using a metric called the "random factor," which adjusts the distribution of the Regina Pats' goals for and goals against metrics. As discussed in the Poisson justification above, hockey can be highly unpredictable in terms of the frequency with which goals are scored. Rather than attempting to predict the exact timing of goals in a game, we account for increased randomness through the application of our random factor parameter, which we initialize to be 3: $\xi_n \sim \mathcal{N}(\mu, \kappa\sigma^2) \mid \kappa = 3$ (Random Factor).

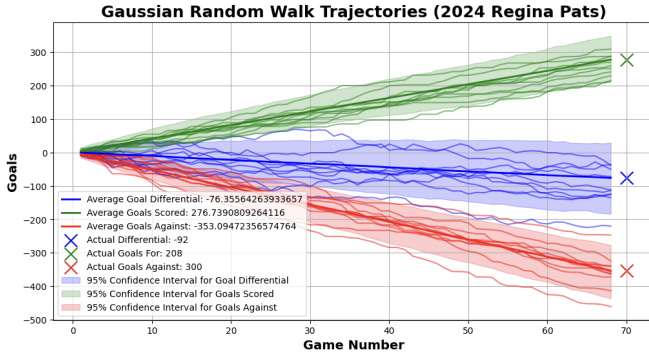


Fig. 2. Random Walk Results for Poisson Distribution: 2024 Regina Pats

The biggest issue with using a Poisson distribution is that when we model wins, the most direct method for doing so is finding all places in a game where the goal differential is positive. For teams like the Regina Pats, that finished near the bottom of the league last season, and programs like the Saskatoon Blades, who won the Eastern Conference, the number of projected wins will be close to 0 and 68 (the length of the regular season after 2017), respectively. By scaling σ with our random factor input for our normal distribution steps, we account for more randomness in the system and improve our win projection. The downside is that the final range of our confidence intervals for goal differential projection expands (Fig 2). A histogram of projected win totals for the Regina Pats and Saskatoon Blades has been included to highlight these discrepancies (Fig 3).

As evidenced by histogram, teams at the top and bottom of the standings can be challenging to predict with a high degree of accuracy, with both the Gaussian step and the Poisson step. There is an evident trade off between predicting wins more effectively and predicting goal differential. When we increase the randomness factor (Fig 5), our ability to predict cumulative goal count with a random walk naturally decreases. However, if we keep it constant at $\kappa = 1$, our mean absolute error for goal differential after 1000 simulations of the WHL's 68-game season on our predictive system sharply decreases (Fig 6):

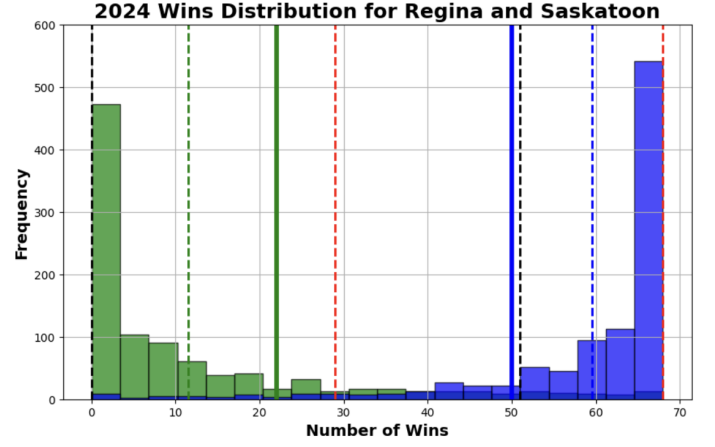


Fig. 3. Gaussian Random Walk Win Distributions: 2024 Regina vs. Saskatoon

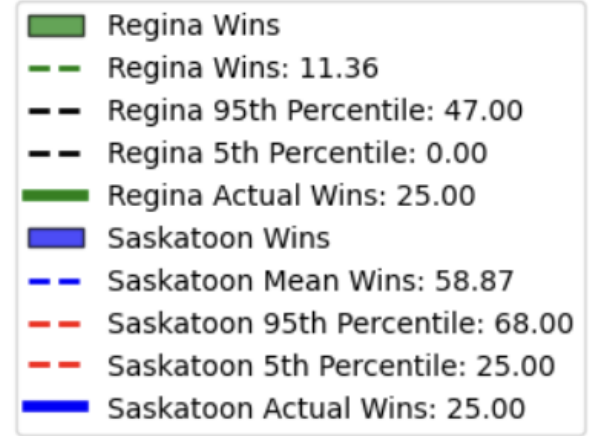


Fig. 4. Legend for Random Walk Win Distributions

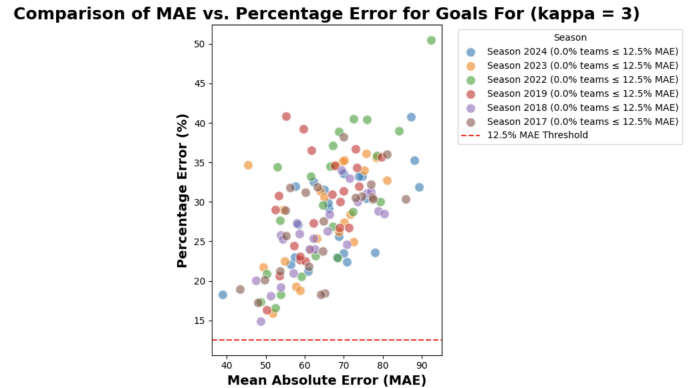


Fig. 5. MAE on Gaussian Random Walk simulation with $\kappa = 3$. All seasons.

Comparison of MAE vs. Percentage Error for Goals For ($\kappa = 1$)

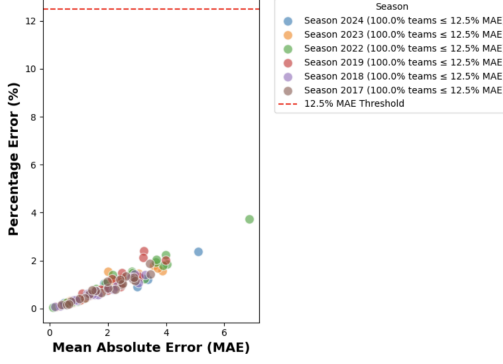


Fig. 6. MAE on Gaussian Random Walk simulation with $\kappa = 1$. All seasons.

IV. RANDOM WALK WEIGHTED AVERAGE MODEL

Our rudimentary model only incorporated one team's average statistics when making decisions regarding their cumulative win and goal differential totals. While it provided us with a strong foundation, it lacked any metric that could be used to compare two teams against one another. In our second Random Walk model, we determine average goals and average goals against with the following equation for team i and team j . Let ω be the weight dictating the weighted average allocation, G_K be the average goals per game for team k , A_K be the average goals against for team k , where $[i, j] \in K$:

$$\begin{aligned} G'_i &= \omega G_i + (1 - \omega) A_k \\ A'_i &= \omega A_i + (1 - \omega) G_k \end{aligned}$$

This pair of equations for team i allows us to iteratively adjust the parameters for team i by leveraging their opponent's statistics. We calculated each team's record against every other team, finding the number of games each team i played against team j over the course of the seasons. We then simulated that number of matchups for each team, ensuring that the sum of post-2017 matchups is equal to the total number of games in the full regular season.

For the 2024 season, using a random factor of $\kappa = 3$ for a Gaussian step produced conflicting results. By cross-referencing with Figure 5, we can conclude that our percentage error decreases when the weight is optimally set to match team i 's statistics. However, there is still too much variance, even with a reproducible seed, to draw any conclusive insights about whether incorporating the statistics of team j improves our projections for team i (Fig 7). Running additional simulations may provide more reliable results, but the key takeaway is that we need a better model.

2024: WHL Average MAE for Goals For and Goals Against

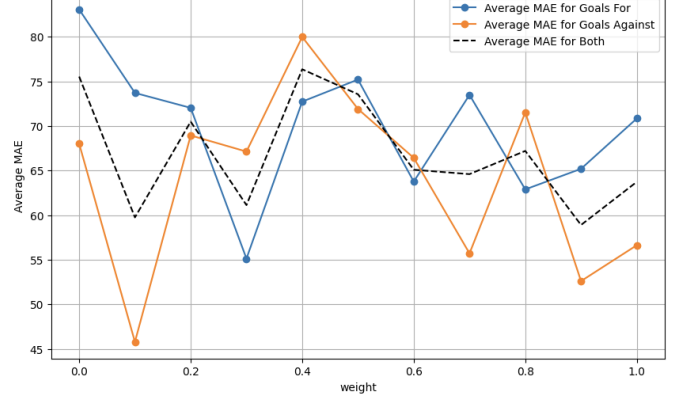


Fig. 7. MAE on Gaussian Random Walk simulation with $\kappa = 3$ (while varying the weighted average composition) for the 2024 season.

V. ELEVATING OUR STAKES: MODELING THE FRONT OFFICE

We will attempt to tackle a much more difficult and nuanced task - that of evaluating and predicting trade success. We will assess the impact of trades on regular season performance, and our datasets will include: (a) game-by-game player and goalie stats, (b) all WHL transactions, (c) cumulative player stats, and (d) cumulative goalie stats. For each team t in each season i , we will develop a metric X_{it} that quantifies how strong player X is during season i on team t , and a metric T_i that quantifies how talented team t is during season i .

VI. SEASON BREAKDOWN

Let n represent the number of trades a team makes during the season. Let I_x be an indicator variable for each player X , where:

$$I_x = \begin{cases} 1 & \text{if player } X_{it} \text{ was on the roster at time } t \\ 0 & \text{if player } X_{it} \text{ was not on the roster at time } t \end{cases}$$

For each team making n trades, we will evaluate the strength of said team in the season on a per-game basis, ultimately storing and tracking the evolution of their team's rating across the whole year. However, with the integration of our trade database, we are effectively able to separate these ratings into blocks. Each of these blocks corresponds to a had a distinct roster - that is, that team $A_{t=3}$ has an entirely different roster than the rest of team A_t , where $t \in R \neq 3$. Specifically:

- **Block 1:** Team's performance from the start of the season until the first trade.
- **Block 2:** Team's performance from the first trade until the second trade.
- And so on, until the trade deadline.

Within each block, we will calculate per-game statistics to quantify the impact of each player on the team's overall performance. We will then aggregate these statistics to form a

cumulative metric of trade exit and entry value, which allows us to evaluate how each trade may be expected to predict future team outcomes.

VII. PERFORMANCE METRICS

To quantify the impact of player trades, we will use the following individual and team statistics in terms of X and T :

A. Player-Level Stats

- **Points per Game (Player PPG):** The total points (goals + assists) scored by player X per game. This metric will allow us to quantify how a player's offensive contributions affect the team's performance.

$$\text{Player PPG}_{X_{it}} = \frac{\text{Goals}_{X_{it}} + \text{Assists}_{X_{it}}}{\text{Games Played by Player } X_{it}}$$

- **Goals:** The total number of goals scored by player X .
- **Assists:** The total number of assists by player X .
- **Penalty Minutes** The total number of assists penalty minutes incurred by player X .
- **Save Percentage (for Goalies):** The total number of saves made by a goalie divided by the total number of shots faced by a goalie.
- **Saves (made by Goalies):** The total number of saves made by a goalie (a stat that adjusts and rewards goalies properly for large volume).

B. Team-Level Stats

- **Points per Game (PPG):** The total points earned by the team (wins and losses) divided by the number of games played:

$$PPG_t = \frac{\text{Total Points Earned by Team } t}{\text{Games Played by Team } t}$$

- **Wins:** The number of games won by the team.
- **Losses:** The number of games lost by the team in regulation.
- **Overtime (OT) Losses/Shootout Losses:** The number of games lost in overtime or a shootout.
- **Goals Scored:** The total number of goals scored by the team in each game.
- **Goals Allowed:** The total number of goals allowed by the team in each game.

These stats will allow us to build a comprehensive picture of how player and team performances evolve before and after trades, with a focus on how individual contributions (goals, assists, saves) impact overall team success (wins, losses, goals scored, points).

VIII. BASE LEVEL IMPACT OF TRADES

Let P_t denote the points per game of team t before and after a trade. The point system used in the WHL is:

Win = 2 pts, Overtime/Shootout Loss = 1 pt, Loss = 0 pts

Let P_{it}^{before} and P_{it}^{after} represent the points per game for team t before and after each trade involving player X during

season i . The goal is to model how trades change the team's performance in terms of points per game.

For a player X , the impact of that player on the team's performance over the last 20 games can be quantified by:

$$\Delta P_{X_{it}} = P_{it}^{\text{after}} - P_{it}^{\text{before}}$$

where:

- $\Delta P_{X_{it}}$ represents the change in points per game due to the addition (or removal) of player X .
- P_{it}^{before} is the team's performance before the trade (with player X).
- P_{it}^{after} is the team's performance after the trade (with player X).

This calculation will be repeated for each traded player X , allowing us to evaluate the total effect of all trades on team performance.

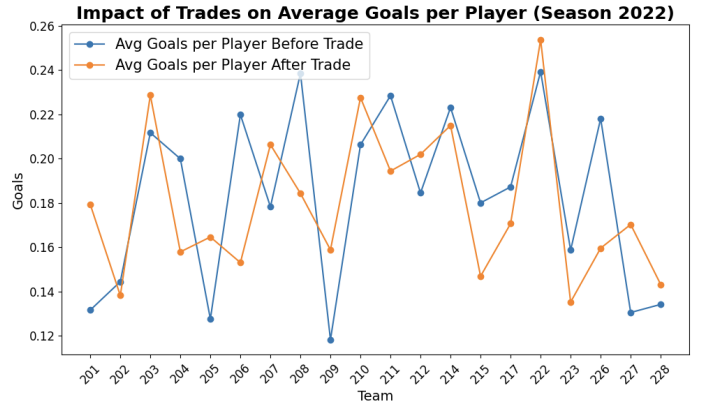


Fig. 8. Team-by-team trade impact of Goals per game (per player)

Our analysis across all years of data bears similar results to shown above. However, looking only at 2022 here to tell our results (on the base level). It is *very* ambiguous as to whether trading for players always increases your teams average goals per game. This generally makes sense - our base model includes both trades here where teams are trying to build their current roster and trades where teams may be selling their best current players for future draft picks. This apparent non-optimizing present strategy will greatly feed the development of our more complicated trade prediction model. Next, we will continue with our simple approach to see the impact on wins, rather than goals, in the 2022 season.

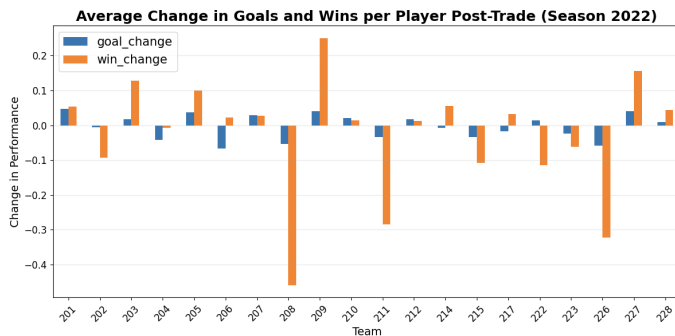


Fig. 9. Team-by-team trade impact of Wins and Goals per pame (per player)

We posit that wins tell us more about the overall trade landscape than goals. A small majority of entries demonstrate that trades, on net, earned more wins per player in the 2022 season. This can likely be attributed to an assimilation of the degrees of two competing strategies. First, the majority of teams are trying to build their present rosters and improve their likelihood of winning now. It is generally difficult to increase and build wins this way. A smaller portion of teams are trying to sell of their roster, abandoning hope for the current season and looking to improve their future prospects only. For these teams, we see that the front office can have a greater degree of impact on making their team worse than the front offices trying to improve their own teams.

We have illustrated that to take a more accurate predictive view, we cannot simply look at the impact of trades across the league. The non present-optimizing behavior by league front offices presents a unique complexity to the overall impact of trades. Thus, we must take a more systematic approach - which inherently carries even greater risk.

IX. A SYSTEMATIC APPROACH TO QUANTIFY FRONT OFFICE TRADES TO PREDICT TEAM PERFORMANCE

Our most ambitious target when taking on this topic was to see whether we could accurately determine the value of trades. We wanted to put ourselves in the shoes of a WHL General Manager (GM) and ask how could we evaluate players we wanted to acquire or deal away via trades?

A. Player Evaluation

A player's ultimate evaluation point will be their plus-minus. Rather than blindly assign them a rating, we wanted to deterministically focus on process. We decided to carefully select their inherent stats and run a logistic regression to determine prominence in determining their $+/-$ ratings. To avoid any correlation between stats: we have taken 3 distinct metrics that each tell a story. First - goals, which appears self evident. Second, assists, which in a way, demonstrates how a player's ability on the ice can elevate the play of those around him. Finally, penalty minutes - something that players look to minimize, as taking penalties is detrimental to team success.

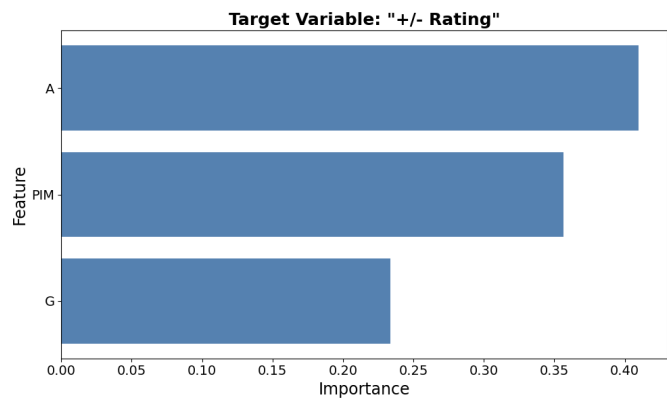


Fig. 10. Logistic regression of key statistic weight coefficients

Assists are the most important factor in determining a player's plus-minus. This is also, in part, to a maximum of two assists being awarded for every goal. Perhaps there is a synergistic factor to assists as discussed prior. Penalty minutes is second - which emphasizes the importance of avoiding the power play for team success. Finally, goals appears to be the least important of the three.

Now, we turn to the other player of interest for GMs - goalies. Goalies are responsible for protecting their teams net and stopping the opposing team's shots. Therefore, assists, penalty minutes, and goals are not requisite evaluation metrics for them. Further, team plus-minus also does not act as a proper target variable. We instead, track to total points earned, with a similar process oriented approach in building a formula with statistics weighted by coefficients derived from a logistic regression. We selected the statistics of both save percentage (which is typically a great indicator for how skilled a goalie is) and total saves. Total saves here, though perhaps correlated to save percentage, is key to realize the value of a great goalie on a bad team. For example, a goalie on a bad team may face twice as many shots as an average goalie, making it harder for him to maintain such a strong save percentage. Therefore, the total saves metric allows for necessary volume recognition.

We have now defined two formulas to define the value of both goalies and players:

$$Player_Rating = .41A + .35P + .24G$$

$$Goalie_Rating = .62Save_Percent + .38Saves$$

Finally, we standardize these ratings using a python Min-MaxScaler, as well as applying necessary weight coefficients to value the best goalie at about 60% of the best skating player - mimicking the job of a GM in identifying and rating players. Now, we turn to team evaluations as the next step to our predictive model.

B. Team Ranking

As explained earlier, within each season, we are going to have teams ranked continuously. Meaning, we are going to

store team ranking for each game. Therefore, our next order of business is how to actually determine these ratings.

We saw very little motivation to depart from the accepted method of an ELO rating. Using our scraped head-to-head data, we were able to assign hockey-adjusted ELO's for **every single team across every single matchup across every single season**. Hockey adjusted - meaning that OT losses would penalize a team less, and OT wins would reward a team less. We knew taking this scope carried risk, but wanted to build the most comprehensive model.

ELO is a rating system where each team's rating changes after a match based on the difference between their expected outcome and the actual result. If a lower-rated team beats a higher-rated one, they gain more points, while the higher-rated team loses points accordingly. The equation to determine how a team's ELO updates is therefore as follows:

$$R' = R + K \cdot \left(S - \frac{1}{1 + 10^{(R_{\text{opp}} - R)/400}} \right),$$

where:

- $R' \rightarrow \mathcal{R}'$ (New ELO Rating)
- $R \rightarrow \mathcal{R}$ (Current ELO Rating)
- $K \rightarrow$ Weight factor (K-Factor)
- $S \rightarrow$ Actual match result

C. Trade-Influenced ELO Progressions

We now have an exciting prospect: a fully-built-out ELO system ranking every team at every moment of a season. As explained earlier, this allows us to distinctly block portions of a season with specific rosters. As these rosters change in-season by trades, we can see how difference in performance is changed by trades. As such, we can see how different trades directly changed performance. For example, let's look at Saskatoon's 2023 season.

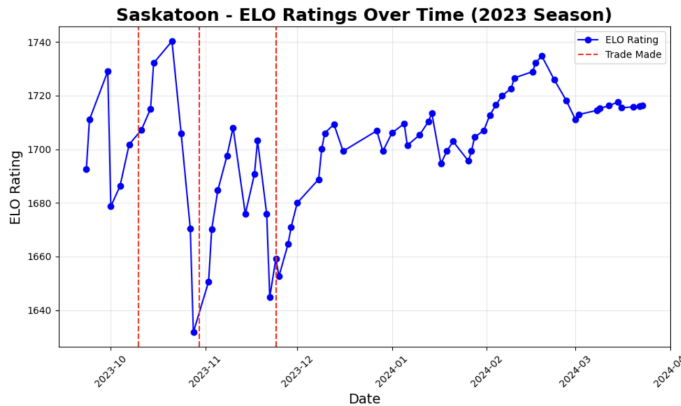


Fig. 11. Saskatoon's 2023 ELO Rating with Trade Indicators

We see that Saskatoon's ELO rating begins with significant volatility. This makes sense - as the teams are starting earlier in the season, there are less data points to judge their overall rating on. Further, we see that there were three trades made

by Saskatoon in the season. The first trade saw them go on great streak. However, after 7 or 8 games, their ELO rating had fallen back down significantly. Then, they made another trade, which led to another significant winning streak. The team started to struggle once more, prompting a third and final trade for the season. This trade saw another leap in their ELO score, followed by a stabilization in the back third of the season. The Saskatoon 2023 season is a great case study into effective managing, all three trades led to inflection points in team performance.

However, others were not so straightforward. For example, let's briefly look at the Regina Pat's 2023 season progression.

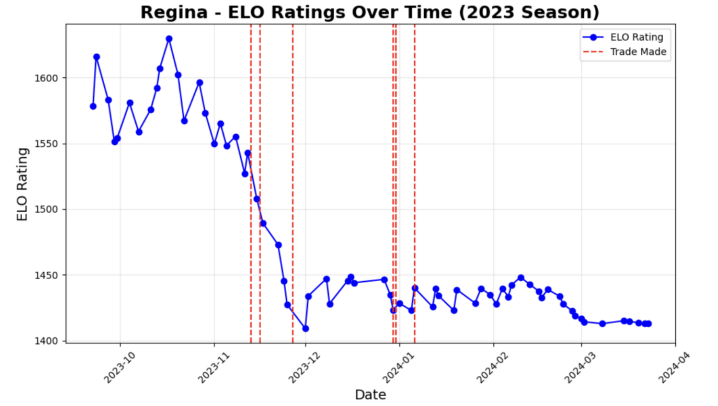


Fig. 12. Regina's 2023 ELO Rating with Trade Indicators

Regina remained very active movers, making six different trades. However, every single one of these moves resulted in a decreasing ELO rating (on average) over the next 5 games. This could indicate two things - first, that the front office was genuinely struggling to add roster pieces to boost current team success in search for a championship. Secondly, that they could have been intentionally disposing of their players to improve their future.

Let's try and build a model and see if it can handle both motivations - trading to improve your team and trading to improve your team's future - at the same time.

D. Predicting Trade Success

We have established an entire rating system for every player across the WHL - each player having a different rating for season. We have also established a continuous ELO rating system for every team, with clear team rating differences caused by trades. Now, we will attempt to merge our various sets and build a logical prediction model for the impact that any trade will have on your team's performance.

Firstly, we want to determine the difference in the assets being traded. Thankfully, we have a fairly simple mechanism to do this. We are able to leverage the already established player rankings and take the difference between the sum of the rating of players received versus given away.

$$Paper_Value = \sum Ratings_{Received} - Ratings_{Given}$$

Each team was then assigned their respective paper value, with the paper value of every trade properly summing to 0. One such limitation of our model is our lack of quantifying draft assets that are traded. We do feel that, since we are also trying to determine how a team trying to sell their assets will effectively worsen their present team, that this shortfall is largely acceptable. However, we note and realize that an even more built model would have made attempts to quantify draft assets.

After each team is assigned their net paper value (the change in value, on paper, of the player assets that comprise their team), we must look at how paper reconciled to on-ice performance.

Because of our continuous build of ELO ratings for every team across the entire season, we were able to pinpoint just how "good" a team was at the time just prior to a trade. We use this as our *before* rating. Our *after* rating, on the other hand, we calculate as the average of the 5 ELO ratings immediately following the trade.

$$Performance_Change = Mean(ELO_{5_After}) - ELO_{Before}$$

The reason we use 5 games, as opposed to the immediate game, is to allow the team some time to mesh together and build chemistry with each other. Admittedly, the selection of 5 was not driven by any analytics, but rather by anecdotal experience. A more detailed analysis could have examined the derivative of the ELO and determine a set optimal number of games to use as the point of the average ELO rating.

We now have the net paper value and performance change of every trade, across all interest seasons, in the WHL. Now, each net paper value is paired with the actual performance change of the team following the trade. In theory, trades with a negative net paper value *should* lead to decreases in actual performance change (and vice versa). Let's put this to the test and predict.

We will now run a simple linear regression. We have effectively paired all trades by now with their intrinsic value (net paper value) and their actual impact (performance change). We will then use each trades net paper value as the independent variable, which will impact (and hopefully predict) the dependent variable (performance change).

If there were any indicator that a modeling course should be required for all math concentrators, the failure of this linear regression would be it. We trained our model on the 2018, 2019, and 2022 trade data. Again, we omitted 2020 and 2021 due to COVID effects voiding most of the season. The training set did not fit very well from the start... with a deterministically low R^2 below 0.2. However, our validation and test sets performed even worse. Our validation

R^2 , which validated against the 2023 season trade set, was **negative**. The test R^2 , which tests against the 2024 trade set, was **also negative**. This was a fairly troublesome result. It is very possible that ranking players is incredible hard to do, and professional GM (such as Saskatoon's) simply do a better job than we do. In theory, positive net paper value should predict an increase in team performance... but this did not work for our validation and test set. Therefore, it is very likely our linear regression model fell victim to severe over-fitting. While more rare in linear regression, we believe this to be due to multicollinearity, among other factors.

Determined, we pivoted towards a Random Forest Regressor model. We structured and paired the trade date the same way. Our results were more promising here on the training set. While not very high, we do feel that this was strong enough to examine the validating and test results.

Looking at the mean absolute percentage error (MAPE) for both the validation and training sets, we see:

- Validation MAPE (2023): 8.6%
- Test MAPE (2024): 17.2%

When looking at why the random forest regressor performed so much better than the linear regression, we must first realize that perhaps the relationship is more non-linear than we realized. After all, trading for the very best player in the league may yield out-sized returns that a linear relationship could not effectively capture. Secondly, it also was more robust to over-fitting (by the nature of the model averaging multiple decision trees). This certainly handheld potential irrelevant features and multicollinearity better.

Our random forest regressor is far from perfect, however. Looking at our results graphically:

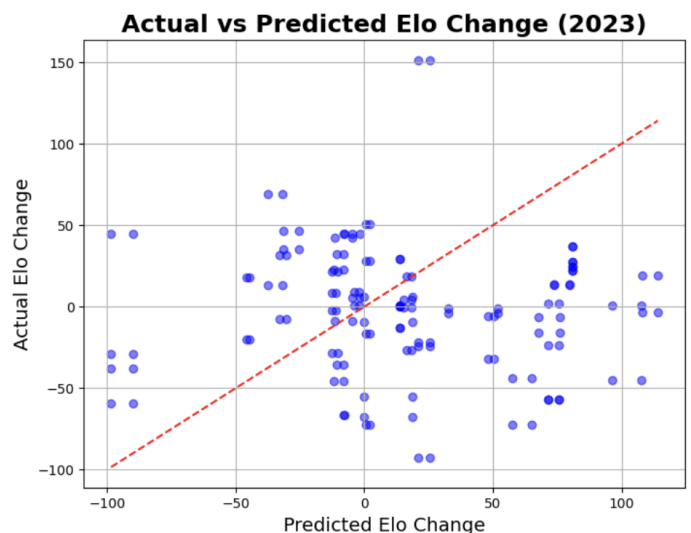


Fig. 13. Predicted vs Actual 2023 ELO Change based on In-Season Trades

This relationship looks generally sporadic. Our domain seemed to be larger than our range, with 2 outliers. This may mean that our model is perhaps too predictive in magnitude, and could benefit from a scaling back. Adding additional

layers of complexity, such as player roles and team context, could potentially increase accuracy.

X. CONCLUSION

In this paper, we have discussed some of the shortfalls and complications of the models throughout, but our work still concluded several key results. We aimed to model regular season wins in the Western Hockey League by examining the impact of trades during specific windows of a team's season. We developed two random walk formulations: one using a Gaussian step and the other a Poisson step. These initial models revealed that goal differential alone was insufficient for predicting a WHL team's regular season performance. To refine the approach, we applied a weighted average of competing teams' performance metrics but found the results inconclusive.

Further, we took on even greater risk and ambition in attempting to tackle the trade problem via player performance and team evaluations, and at the end have yielded mixed results. The prediction power was strong in terms of error; however, graphically we clearly saw that our predicted ELO change based solely on the net paper value of the players exchanged in the trades is not the most consistent predictor of actual ELO change. Moving forward, a more robust framework is needed—one that captures the dynamism of roster changes, particularly the influence of trades.

APPENDIX A PYTHON NOTEBOOKS

- 1) Primary Notebook: Link to Notebook
- 2) Supplemental Notebook for Data Scraping: <https://github.com/yourusername/notebook>

APPENDIX B KAGGLE, FUTURE MODELS, AND IMPLICATIONS

All of our .csv and .pkl files have been uploaded as a public dataset to Kaggle. We hope that others will continue to model the WHL in further depth. We feel strongly that better predictive models could improve not only team success, but player development as whole. Quantifying trade impact would give players a new lens through which they can assess their impact on their program: Kaggle Link.

APPENDIX C DATA SCRAPING

We scraped data from the following websites, which we have cited in our references section as well. Item 1 has a feature where a user can generate data manually at request. It is an incredibly helpful tool.

- 1) WHL Media Access
- 2) WHL Transaction Log

APPENDIX D HOCKEY STATISTICS

- 1) Goals: The total number of goals a player scores during a game or season. A goal is counted when the puck crosses the opponent's goal line.
- 2) Assists: The number of times a player helps set up a goal by passing the puck to the goal scorer. Typically, up to two assists can be awarded per goal.
- 3) Points: The sum of a player's goals and assists. Points are a primary measure of offensive contribution.
- 4) Plus/Minus: A statistic that tracks the goal differential while a player is on the ice. A player earns a +1 for a goal scored by their team and a -1 for a goal scored by the opposing team.
- 5) Goals Against Average (GAA): A goalie statistic that calculates the average number of goals allowed per game. It is calculated by dividing the total number of goals allowed by the number of games played and multiplying by 60.
- 6) Save Percentage (SV%): A goalie statistic that measures the percentage of shots on goal stopped by the goalie. It is calculated by dividing the number of saves by the total number of shots faced.
- 7) Shots on Goal (SOG): The number of shots a player takes that are directed on target to the opposing goalie. A shot that is blocked or missed is not counted.
- 8) Penalty Minutes (PIM): The total time a player spends in the penalty box due to penalties committed. This statistic reflects a player's discipline and impact on special teams.

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Please see the Attribution of Effort PDF also submitted on Canvas for the breakdown of the 75+ hours of effort that the authors put in.

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We would like to give a necessary attribution to ChatGPT, which helped us scrape data from the websites in Appendix B and in random spots in our notebooks. We derived all of the logic and initial code ourselves, but ChatGPT lent us a hand with as we sought to create powerful visuals and debug persistent syntax errors.

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